

CRITICAL IMPLEMENTATION REQUIREMENTS

Non-Regression Requirement (Highest Priority)

This project is an enhancement of an existing production application.

The new Translation module MUST NOT:

- Modify existing routes
- Modify existing APIs
- Break existing pages
- Change existing design language
- Change authentication flow
- Change application layout
- Change state management patterns
- Introduce breaking database changes

Existing modules must continue working exactly as they do today:

- Convert Document
- Chat With Document
- Stats & Results
- Settings

The new Translation module must be completely isolated.

Use:

Feature Module Architecture

Example:

```
src/modules/translation  
  
components/  
services/  
hooks/  
pages/  
types/  
api/
```

No existing modules should require modification other than:

Header Navigation
New Route Registration
Shared Theme Registration

All existing functionality must pass regression testing before merge.

Agentic AI Architecture

Do NOT implement translation using a single LLM call.

Implement a CrewAI-based multi-agent workflow.

Goal:

Higher translation quality

Better formatting preservation

Better glossary enforcement

Better explainability

Agent 1

Translation Coordinator Agent

Responsibility:

- Receive job
- Validate configuration
- Build workflow
- Coordinate all agents

Recommended Model:

Gemini 2.5 Flash

Reason:

Fast Cheap Large Context

Agent 2

Document Analysis Agent

Responsibility:

- Detect language
- Identify document structure
- Identify tables
- Identify charts
- Identify speaker notes
- Identify formulas

Output:

```
{
  "documentType": "",
  "language": "",
  "sections": []
}
```

Recommended Model:

Gemini 2.5 Flash

Agent 3

Glossary Enforcement Agent

Responsibility:

- Load glossary
- Identify protected terms
- Create translation rules

Example:

```
NOVATHERA → NOVATHERA

TradeIQ → TradeIQ
```

Recommended Model:

Groq
Llama 4 Scout

Agent 4

Translation Agent

Responsibility:

- Translate chunks
- Apply translation mode
- Apply glossary

Recommended Models:

Priority:

1. Gemini 2.5 Flash
2. OpenRouter DeepSeek
3. Mistral Large

Use configurable provider routing.

Agent 5

Formatting Preservation Agent

Responsibility:

- Verify formatting
- Verify table structure
- Verify chart labels
- Verify notes

Recommended Model:

Groq
DeepSeek

Agent 6

Quality Review Agent

Responsibility:

- Compare original vs translated
- Score translation quality
- Detect omissions
- Detect hallucinations

Output:

```
{  
  "quality":96,  
  "issues":[]  
}
```

Recommended Model:

Gemini 2.5 Flash

Agent 7

Report Generation Agent

Responsibility:

- Generate Translation Report
- Generate Comparison Report
- Generate Audit Report

Recommended Model:

Groq
Llama

CrewAI Flow

Translation Coordinator

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Document Analysis

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Glossary Enforcement

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Translation

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Formatting Verification

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Quality Review

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Report Generation

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Persist Results

Model Selection Strategy

Default Cost Optimized Setup

Document Analysis

Provider:

Gemini

Model:

gemini-2.5-flash

Translation

Provider:

Gemini

Model:

gemini-2.5-flash

Fallback:

OpenRouter

Quality Review

Provider:

Gemini

Model:

gemini-2.5-flash

Formatting Verification

Provider:

Groq

Model:

llama-4-scout

Free / Low Cost Models Priority List

OpenRouter

Preferred:

openrouter/free

Then:

DeepSeek

Qwen

Nemotron Free Models

Poolside Laguna

Use OpenRouter free routing whenever possible.

Groq

Preferred:

Llama 4 Scout

Llama 4 Maverick

DeepSeek Distill

GPT OSS

Available through Groq hosted model catalog.

Gemini

Preferred:

gemini-2.5-flash

gemini-2.5-flash-lite

These are Google's fastest and most cost-efficient options.

Mistral

Preferred:

mistral-large

mistral-medium

ministral-8b

Bedrock

Use only for:

Enterprise Override

High Quality Review

Large Batch Processing

Recommended:

Claude Sonnet

Nova Pro

Agent Failure Handling

If an agent fails:

Retry 3 Times

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Switch Provider

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Continue Workflow

Example:

Gemini Failure

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OpenRouter DeepSeek

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Groq Llama

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Raise Error

Expected Outcome

The Translation Module should:

- Preserve existing application behavior

- Require zero regression fixes
- Use agentic orchestration
- Minimize token cost
- Maximize translation quality
- Support enterprise document workflows
- Support future model additions without redesign